

FORECAST DECISION DASHBOARD



FORECAST DECISION

[Pilot Deployment]

Confidence

LOW

Business Risk

LOW

Deployment Strategy

GLOBAL

Recommended Model

V2_9_Prophet

Next Action

Pilot for 4 weeks before enterprise rollout.



Best WAPE	Weekly Win Rate	Volume CV	Models Evaluated
9.47%	51.52%	7.59%	92

Executive Takeaway

Machine Learning currently ranks first, however statistical evidence is insufficient to justify immediate production rollout.

Key Evidence

- Proposed model achieves 9.47% WAPE.
- No incumbent baseline is available for comparative evaluation (Greenfield).
- Model V2_9_Prophet ranked #1 out of 92 candidates.
- Volume remains stable (CV: 7.59%), creating a favorable forecasting environment.

Accuracy Assessment (Incumbent vs ML)

Business Question: *Did human planners or machine learning algorithms produce more accurate forecasts?*

Observation: No incumbent baseline is available for comparison (Greenfield scenario). **Evidence:** Baseline Availability: False **Conclusion:** No incumbent baseline is available for comparison (Greenfield). **Decision Support:** The ML model establishes the initial operational baseline for future comparison.

Model Champion Selection

Business Question: Which forecasting model ranked first across all evaluated candidates?

Observation: Model V2_9_Prophet ranked #1 across 92 scored candidates with a composite score of 75.32/100. It leads V3_33_Prophet by 7.22 points.

Primary Evidence: - Winsorized Composite Score: 75.32/100 - WAPE: 9.47% - Weekly Win Rate: 51.52%

Supporting Evidence: - Absolute Bias: 0.83% - IQR Stability: 13.85% - Hit10 Rate: 65.66% - Runner-up Composite Score: 68.09/100 - Total Models Scored: 92

Conclusion: Model V2_9_Prophet achieved the highest composite score across all evaluated candidates.

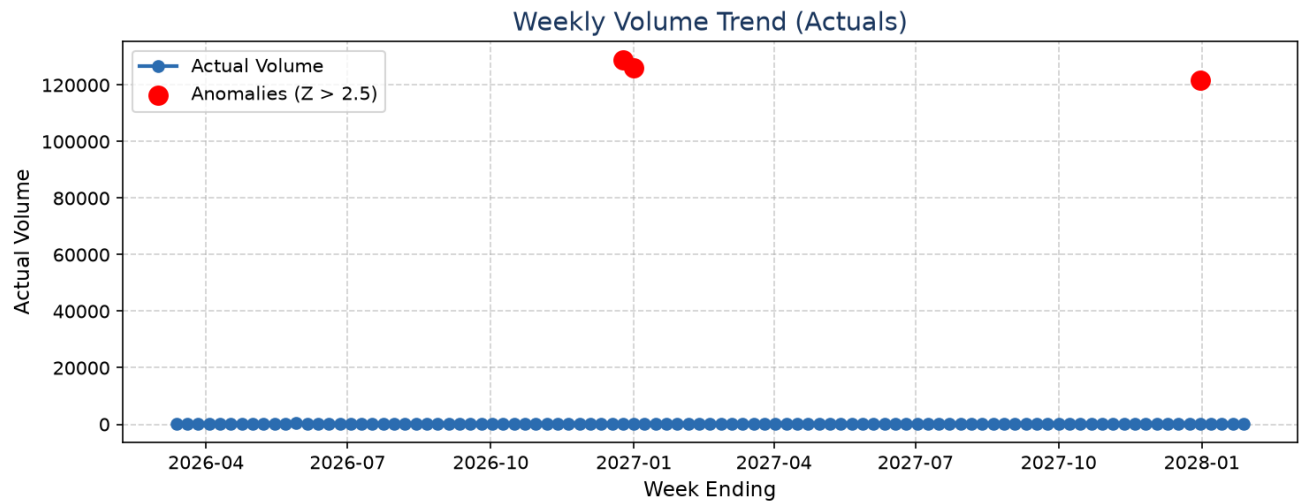
Decision Support: If confidence is Low, the champion ranking is reported for informational purposes only.

(Chart Placeholder: [horizontal_bar_chart] Model Composite Scores - Ranking of all evaluated models based on Winsorized Composite Score.)

Appendix A: Full Model Scorecard Breakdown

Business Context (Actuals)

Charts



Business Question: Was the underlying demand volume stable enough to be forecasted?

Observation: Demand variability remained low (CV 7.59%). 3 statistical anomalies were detected ($Z > 2.5$). Overall volume exhibits a Flat trend.

Primary Evidence: - Coefficient of Variation: 7.59% - Anomaly Count: 3 - Trend Direction: Flat

Supporting Evidence: - Mean Volume: 188,435

Conclusion: Demand remained statistically stable throughout the evaluation period.

Decision Support: Volume conditions should be considered favorable when reviewing forecast accuracy results.

Forecast Degradation Analysis

Business Question: *Did unexpected volume spikes or drops cause the forecast to fail?*

Observation: Detected 3 volume anomalies, but none coincided with severe forecast degradation.

Primary Evidence: - Total Anomalies: 3 - Severe Degradation Events: 0

Conclusion: The champion model successfully maintained accuracy despite anomalous volume events.

Decision Support: The evaluated model demonstrated resilience against external volume shocks.

Appendix B: Detailed Event Timeline

Appendix A: Statistical & Data Audit

Business Question: What are the underlying statistical proofs for this document?

Conclusion: See attached tables for full statistical, normalization, and exclusion audit.

Model	Composite Score	WAPE	Abs Bias	Stability IQR	Hit10
V2_9_Prophet	75.32	9.47%	0.83%	13.85%	65.66%
V3_33_Prophet	68.09	10.34%	2.73%	16.11%	58.59%
V2_2_ARIMA	67.56	11.57%	2.96%	15.35%	59.60%
V2_4_Prophet	62.73	11.81%	3.85%	17.76%	54.55%
V2_37_Prophet	60.55	9.50%	3.92%	20.80%	48.82%
V2_38_Prophet	57.67	11.05%	0.37%	25.40%	44.44%
V3_38_Prophet	56.53	13.02%	1.78%	21.61%	43.77%
V3_22_Prophet	55.80	14.46%	1.16%	21.89%	44.44%
V3_0_LR_With_VI	55.65	12.15%	7.46%	18.65%	48.48%
V2_32_Prophet	53.06	15.67%	3.12%	21.88%	45.45%
V3_15_Prophet	51.95	13.79%	2.78%	25.30%	42.93%
V3_3_ARIMA	51.20	12.45%	3.70%	26.11%	41.41%
V2_40_Prophet	49.29	13.76%	6.36%	23.03%	41.41%

V3_37_Prophet	46.99	16.87%	5.28%	23.73%	41.92%
V3_48_Prophet	46.24	15.75%	9.39%	22.59%	45.45%
V2_47_Prophet	45.68	17.80%	0.01%	34.05%	38.38%
V3_9_Prophet	45.50	16.58%	1.37%	28.49%	35.86%
V2_22_Prophet	45.34	17.33%	4.90%	26.40%	42.42%
V2_13_Prophet	44.51	19.72%	1.24%	27.04%	38.38%
V3_46_Prophet	44.38	15.91%	2.56%	34.50%	36.87%
V3_19_Prophet	43.84	18.31%	0.22%	30.64%	35.69%
V3_40_Prophet	43.46	19.21%	2.07%	28.13%	38.38%
V3_25_Prophet	42.88	15.61%	4.93%	28.54%	35.86%
V3_27_Prophet	41.94	15.37%	4.92%	32.96%	35.35%
V2_29_Prophet	41.57	15.83%	9.92%	23.64%	37.37%
V2_43_Prophet	41.27	18.84%	2.12%	34.48%	35.35%
V3_6_Prophet	41.24	17.15%	6.76%	26.96%	37.49%
V3_20_Prophet	40.86	14.54%	5.84%	34.01%	33.00%
V3_8_Prophet	40.82	16.00%	8.35%	58.01%	42.42%
V3_44_Prophet	40.81	13.78%	10.87%	24.23%	33.84%

V3_0_XGB	40.22	16.06%	5.11%	31.37%	33.33%
V3_1_Prophet	40.12	15.87%	6.76%	26.70%	31.31%
V3_11_Prophet	39.48	18.67%	6.85%	28.58%	39.73%
V2_6_Prophet	38.88	17.93%	3.19%	58.93%	26.60%
V3_43_Prophet	38.60	15.49%	8.00%	41.19%	35.02%
V3_0_Prophet	38.13	21.18%	0.75%	38.75%	30.14%
V3_35_Prophet	37.36	21.79%	9.91%	28.60%	49.49%
V3_4_Prophet	36.98	18.54%	5.77%	34.33%	33.11%
V2_11_Prophet	36.67	14.85%	13.63%	27.51%	38.38%
V2_28_Prophet	36.64	18.29%	5.83%	40.11%	31.82%
V3_7_Prophet	36.53	20.33%	3.14%	44.62%	24.82%
V3_0_ARIMA	36.43	18.43%	5.14%	41.96%	25.38%
V3_10_Prophet	35.96	19.08%	6.35%	32.13%	33.33%
V3_32_Prophet	35.58	17.86%	10.60%	26.66%	34.34%
V2_24_Prophet	34.88	25.31%	0.78%	42.70%	32.32%
V3_31_Prophet	34.20	22.74%	3.05%	39.95%	26.26%
V3_34_Prophet	33.16	22.35%	4.69%	34.44%	30.51%

V2_44_Prophet	33.09	22.92%	3.98%	38.22%	26.77%
V3_16_Prophet	32.38	22.44%	5.18%	35.34%	27.44%
V3_1_ARIMA	31.97	18.16%	14.67%	28.20%	38.64%
V3_17_Prophet	31.08	23.34%	5.59%	43.11%	25.25%
V2_0_ARIMA	30.77	17.93%	11.29%	52.88%	26.77%
V3_12_Prophet	30.48	20.58%	9.38%	32.63%	31.06%
V3_18_Prophet	29.96	26.96%	3.08%	43.46%	26.77%
V3_10_XGB	28.33	30.10%	1.57%	59.38%	22.22%
V3_23_Prophet	25.58	24.78%	9.64%	40.12%	25.93%
V4_0_LR_With_VI	24.87	25.03%	10.10%	45.33%	26.60%
V3_4_ARIMA	24.45	26.68%	8.86%	52.85%	21.34%
V3_13_Prophet	24.43	29.37%	6.19%	44.79%	25.05%
V2_10_Prophet	24.14	23.57%	15.58%	31.41%	37.88%
V4_10_XGB	23.20	34.52%	2.29%	46.77%	24.24%
V4_11_XGB	22.96	26.65%	10.39%	43.63%	28.28%
V3_14_Prophet	22.90	30.89%	6.21%	47.30%	17.17%
V4_2_ARIMA	22.69	30.36%	6.94%	48.65%	14.14%

V2_23_Prophet	21.13	24.73%	14.13%	37.52%	28.28%
V3_30_Prophet	20.34	28.71%	10.96%	52.14%	22.22%
V2_46_Prophet	17.73	23.55%	21.68%	28.09%	27.27%
V4_0_ARIMA	17.55	51.20%	2.45%	90.99%	20.20%
V3_2_ARIMA	17.52	27.23%	15.25%	45.94%	26.46%
V2_0_LR_With_VI	16.94	30.22%	12.84%	52.16%	20.54%
V4_6_XGB	15.39	61.33%	4.61%	72.75%	18.18%
V3_21_Prophet	14.88	35.67%	9.45%	79.68%	18.86%
V3_24_Prophet	14.06	28.23%	17.71%	36.51%	22.22%
V3_28_Prophet	11.92	29.11%	18.97%	40.84%	27.10%
V3_3_Prophet	9.39	31.68%	20.93%	33.98%	32.56%
V3_0_LR	8.82	42.34%	11.18%	51.37%	23.64%
V2_1_ARIMA	8.38	31.62%	24.19%	70.80%	18.99%
V2_3_ARIMA	6.53	35.72%	17.75%	68.18%	27.27%
V2_3_Prophet	5.18	35.68%	19.14%	49.49%	26.26%
V4_9_XGB	4.29	49.20%	15.71%	69.82%	19.70%
V4_15_XGB	3.46	37.48%	19.06%	57.63%	15.15%

V3_47_Prophet	0.34	53.68%	37.82%	48.02%	30.81%
V2_0_LR	0.00	50.01%	37.21%	54.27%	23.57%
V2_0_Prophet	0.00	41.72%	38.82%	46.09%	26.43%
V2_12_Prophet	0.00	206.76%	206.76%	209.71%	0.00%
V2_16_Prophet	0.00	57.44%	47.96%	121.80%	23.23%
V2_35_Prophet	0.00	53.75%	47.69%	78.90%	20.20%
V2_4_ARIMA	0.00	43.67%	28.10%	63.89%	20.78%
V3_3_XGB	0.00	60.39%	46.88%	110.28%	12.12%
V3_9_XGB	0.00	58.75%	43.51%	81.48%	12.63%
V4_0_LR	0.00	219.57%	180.17%	273.58%	4.04%
V4_1_XGB	0.00	41.36%	26.76%	55.45%	20.20%

Statistical Test	Result
Confidence Level	Low
P-Value (Wilcoxon)	0.5298
Weekly Win Rate	51.52%

Data Quality Metric	Value
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Coefficient of Variation	7.59%
Identified Anomalies	3
Trend Direction	Flat